Temperature and particle motion

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use a particle model to distinguish average motion from total energy.

SUCCESS CRITERIA - I can...

1. Describe temperature in terms of microscopic motion.
2. Use kelvin in proportional temperature comparisons.
3. Avoid treating a single particle speed as the temperature.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

An object at 350 K sits in a vacuum facing surroundings at 230 K. How can it cool, and are absolute-zero surroundings required?

Marzano check - circle: ① ② ③

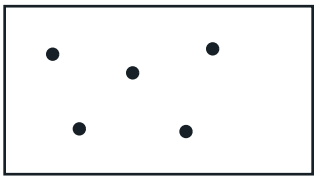
02 Worked example: explain the model

Reference: $K_{avg} \propto T$; $T(K) = T(°C) + 273.15$.
Ideal gas: mean translational kinetic energy per particle scales with absolute temperature. ∝ means proportional to.

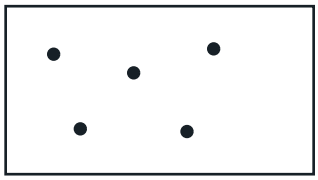
Worked reasoning: In a gas model, warmer temperature corresponds to greater average random molecular kinetic energy. Individual particles still have a range of speeds.

Your turn: Draw two equal-size samples using dots and arrows; label arrows as particle velocities and make the warmer sample's average arrow length greater.

Sample A



Sample B



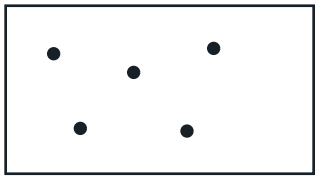
Add velocity arrows; use lengths to show a range of speeds.

Marzano check - circle: ① ② ③

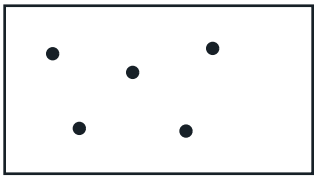
03 Read and interpret the model

Each box represents a sample of the same ideal gas. One is cooler and one warmer. Add particle-velocity arrows with a range of lengths in each box. Make the warmer sample's average kinetic energy greater, without making every particle identical.

Sample A



Sample B



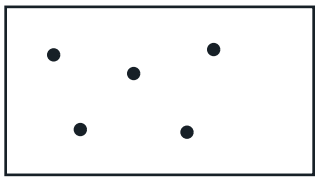
Add velocity arrows; use lengths to show a range of speeds.

Marzano check - circle: ① ② ③

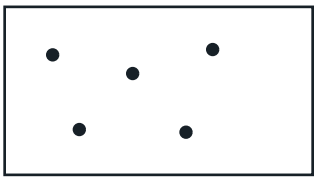
04 Apply the physics

An ideal-gas simulation warms a sample from 300 K to 600 K. Use average translational energy = (3/2)kT with $k = 1.38 \times 10^{-23}$ J/K. Calculate the average energy per particle at each temperature. Label the model with the comparison.

Sample A



Sample B



Add velocity arrows; use lengths to show a range of speeds.

Marzano check - circle: ① ② ③

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05 Explain your reasoning

A student doubles 20°C to 40°C and claims average particle kinetic energy doubles. Explain why the scale matters.

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

A constant 40 N forward force acts at steady 6 m/s for 8 s. Find mechanical power and work.
Use the table to distinguish energy from its transfer rate.

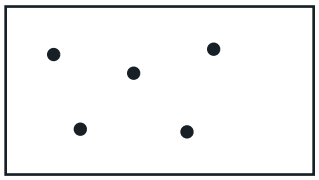
Case	Energy (J)	Time (s)	Power (W)
A	____	____	____
B	____	____	____

Marzano check - circle: ① ② ③

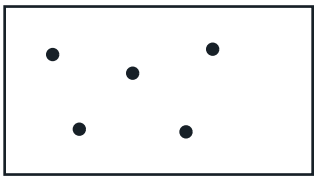
07 Exit check

For a fixed ideal-gas particle mass, average kinetic energy doubles. By what factor does rms speed change?

Sample A



Sample B



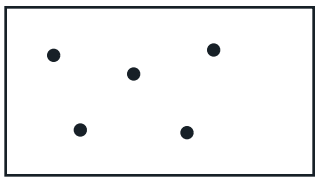
Add velocity arrows; use lengths to show a range of speeds.

Marzano check - circle: ① ② ③

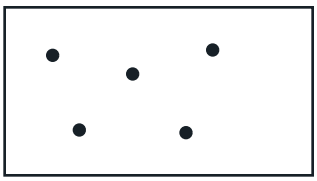
08 STRETCH: test the model

For the same ideal gas, typical molecular speed scales with√T in kelvin 0. Compare typical speeds at 300 K and 1200 K. Why is a Celsius ratio inappropriate?

Sample A



Sample B



Add velocity arrows; use lengths to show a range of speeds.

Marzano check - circle: ① ② ③